

S2S: A deep learning method for the radiological diagnosis of fine-grained diseases spanning screening to subtyping

Highlights

- AI framework integrating screening to subtyping for precise disease diagnosis
- AI system outperforms existing research in detecting and classifying rare tumors
- Novel findings in the human-AI collaboration beyond enhancing physician accuracy
- Framework adaptable to diverse diseases and clinical settings

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In brief

Diagnosing complex diseases like cancers demand nuanced analysis of medical images. This study presents S2S-Med, an AI system that streamlines radiological workflows (from detecting abnormalities to classifying cancer subtypes) using CT scans. Outperforming existing tools, S2S-Med improves diagnostic accuracy while aiding physicians, who achieve better outcomes when collaborating with AI. By integrating multi-stage clinical reasoning into AI, this approach enhances trust and precision in personalized diagnostics, offering a scalable solution for challenging diseases where subtle differences dictate treatment.



Article

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THE BIGGER PICTURE Accurate diagnosis of fine-grained diseases, such as diverse cancer subtypes, is critical for personalized treatment but remains challenging due to subtle radiographic differences and clinician workload. Current AI tools focus narrowly on single tasks, limiting real-world utility. Our screening-to-subtyping (S2S) AI framework bridges this gap by integrating detection, segmentation, and classification into a unified workflow, mimicking clinical reasoning. Following this framework, a practical diagnostic system called S2S-Med is developed, which demonstrates exceptional accuracy in diagnosing 43 mediastinal tumor subtypes, enhances physician decisions, and offers novel insights into human-AI collaboration. This approach reduces diagnostic errors and workload, paving the way for AI-driven precision medicine across diseases. By addressing both technical and clinical complexities, S2S AI could transform radiology, enabling earlier interventions and tailored therapies globally. The observations in the human-AI collaboration may also contribute to new perspectives in clinical practices and foster new paradigms of diagnosis.

SUMMARY

Radiographic images play a critical role in disease diagnosis, but accurately interpreting them requires considerable expertise and workload. Recent research has advanced artificial intelligence-based medical image analysis, but such advancements remain limited in real clinical practice where multistage diagnosis is required for fine-grained diseases. This study proposes a screening-to-subtyping (S2S) AI paradigm specifically designed for accurate radiological diagnosis of fine-grained diseases, encompassing the entire diagnostic process from initial screening to final subtyping. The S2S framework integrates information from multiple diagnostic phases, radiological viewpoints, lesion dimensions, and imaging modalities to address complex diagnostic challenges. Evaluation using a large-scale, multi-center radiography dataset of fine-grained thoracic cancer subtypes demonstrates the system's robust performance. Furthermore, this investigation offers novel insights into human-AI collaboration for diagnosing intricate fine-grained pathologies. Our results highlight the substantial clinical potential of S2S AI across varied healthcare environments and disease entities, facilitating deeper integration of artificial intelligence in radiological diagnostics.

INTRODUCTION

The precision medicine paradigm has highlighted the importance of nuanced treatment strategies tailored to the various subtypes

of fine-grained diseases, such as thoracic cancers.¹ These diseases are characterized by complex heterogeneity, with each subtype requiring a distinct therapeutic approach, which is determined through a sequential diagnostic workflow from screening



to disease subtyping. The emergence of radiological techniques has provided a promising path for physicians to utilize radiographic images for the diagnosis of these fine-grained diseases.^{2,3} However, the detailed interpretation of radiographic images necessitates significant expertise and is labor intensive, resulting in an increased risk of errors, even among experienced physicians.⁴ Consequently, the accurate radiological diagnosis of fine-grained diseases remains a significant unmet clinical challenge, necessitating innovative technological solutions.

In recent decades, artificial intelligence (AI) has emerged as a transformative technology with the potential to revolutionize radiological diagnosis for various diseases.^{5,6} However, existing AI-based systems, including established frameworks like CAIMEN-CL,⁷ often fall short in addressing the complexities of fine-grained diseases due to a lack of holistic nature—the “width” and the “depth”—required for radiological diagnosis in real-world clinical settings. For instance, while CAIMEN-CL demonstrates proficiency in disease segmentation and classification, its limited integration of cross-task and cross-granularity features restricts its accuracy in fine-grained subtype classification. Specifically, width requires comprehensive coverage and high accuracy across all key diagnostic tasks rather than one, which include disease detection, segmentation, and classification. Depth necessitates meticulous attention to the fine-grained subtypes in addition to coarse-grained disease types. The integration of both width and depth is crucial for reducing diagnostic errors and facilitating in-depth disease analysis, which is of significant importance for the development of personalized treatment plans in real-world clinical settings.

Despite the acknowledged importance of these dual dimensions, achieving both presents several challenges. First, distinguishing between fine-grained subtypes is complicated by the high degree of inter-subtype similarity and intra-subtype variability observed in radiographic images. Second, accurate subtyping relies on the precision of the preceding diagnostic stages, such as disease detection and segmentation, which can be affected by the complex anatomical structures surrounding disease locations. Overcoming these challenges is crucial to ensure comprehensive support throughout the diagnostic workflow and to deliver meaningful clinical benefits.

Our study introduces a screening-to-subtyping (S2S) AI aimed at enhancing the accuracy of radiological diagnosis for fine-grained diseases through a comprehensive diagnostic AI workflow. The S2S AI prioritizes a concurrent approach to address both the width and the depth of radiological diagnosis, a strategy that has not been previously highlighted in the literature. On the width dimension, the S2S AI covers the key radiological diagnostic tasks, including disease detection, segmentation, and classification. Additionally, effective methods are introduced to enhance the performance of each individual diagnostic task and to integrate their results, facilitating a lesion-centric diagnosis and improving diagnostic precision. On the depth dimension, the study proposes techniques for extracting multi-scale features of lesions, which are crucial for distinguishing between fine-grained disease subtypes. By addressing both the width (comprehensive task integration) and depth (fine-grained subtype discrimination), our approach advances toward personalized treatment planning while mitigating the risks of diagnostic oversights, a critical step toward translational adoption.

To demonstrate the practical application of the S2S AI, we have developed S2S-Med, an S2S AI system specifically designed for the fine-grained radiological diagnosis of mediastinal neoplasms, a diverse subset of thoracic cancers comprising 43 subtypes. S2S-Med was developed and evaluated using a real-world, large-scale, multi-center dataset of computed tomography (CT) scans.⁷ The system has demonstrated high accuracy throughout various diagnostic stages, with particularly impressive performance in differentiating mediastinal neoplasm subtypes using unenhanced CT scans, a task that poses significant challenges for physicians. Moreover, S2S-Med has shown a commendable ability to avoid missed diagnoses in high-risk cases, underscoring its clinical utility in providing precise and comprehensive diagnostic insights.

Furthermore, to assess the clinical significance of S2S-Med system, we conducted an experiment involving physicians with varying levels of experience, who provided diagnoses both independently and with the assistance of AI. The results indicate that S2S-Med offers significant, experience-independent, and trust-related support to physicians. Importantly, the diagnostic accuracy of the physicians, even when they were aided by AI, did not exceed that of S2S-Med for these complex cancer subtypes. This finding suggests the necessity of reevaluating the human-AI collaboration paradigm in the radiological diagnosis of complex fine-grained diseases.

RESULTS

Concept and architecture of S2S AI

S2S AI is specifically designed for the radiological diagnostic workflow of fine-grained diseases. It stands apart from conventional manual procedures, which place a significant burden on expert physicians, and from existing AI-based techniques, which are typically confined to single diagnostic tasks or coarse-grained disease categories (see [Figure 1](#)).

S2S AI incorporates several techniques to address the width and depth dimensions of radiological diagnosis, mirroring the complexities of clinical diagnostic practice (see [Figure 2](#)). Regarding the depth dimension, we noted that the distinguishing features among different disease subtypes in radiographic images are characterized by multi-scale attributes. These include global-scale features (e.g., the location of lesions within radiographs), local-scale features (e.g., the distribution of image contrast within lesions), and three-dimensional (3D)-scale features (e.g., lesion-specific variations across different slices of 3D radiographic data). In response to these observations, we developed two specialized modules for the purpose of fine-grained radiological disease diagnosis. The first module, referred to as the cross-scale module, is crafted to extract both global-scale and local-scale lesion features, thereby capturing two-dimensional (2D) lesion information within a single radiograph slice. The second module, called the cross-slice module, is designed to integrate the 2D lesion information across multiple radiograph slices to construct a comprehensive 3D lesion profile. By employing these dual modules, the S2S AI is capable of extracting a comprehensive view of lesion characteristics across various scales, thereby facilitating precise and detailed radiological diagnosis of fine-grained diseases.

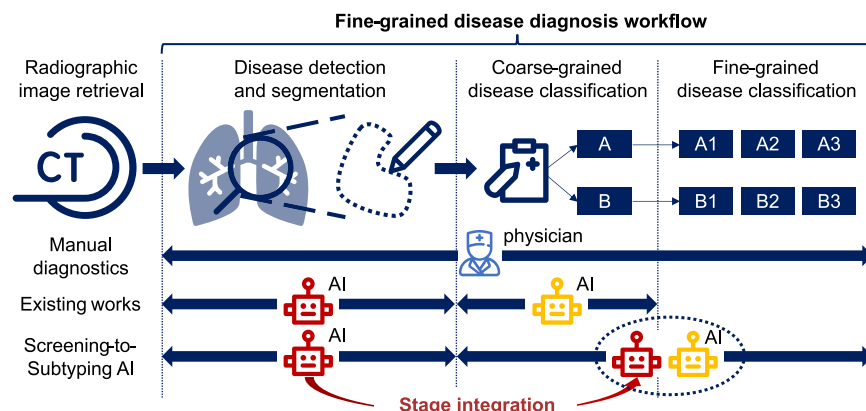


Figure 1. Concept of S2S AI

We compare the diagnostic coverage and responsible roles among conventional manual diagnostics, existing AI works, and our S2S AI.

Through a data selection pipeline (see Figure 3), we gathered 687 contrast CT scans and 4,094 plain CT scans from the development centers, as well as 1,126 plain CT scans from the testing centers.

For the disease detection and segmentation tasks, we included 687 contrast CT scans and 1,106 plain CT scans from the development centers, along with 264

To address the width dimension, it is necessary to enhance the performance of the S2S AI across individual diagnostic tasks and to strengthen the connections between these tasks. To this end, three modules were incorporated into the S2S AI. First, a cross-view module is introduced to simulate the radiograph reading strategy of expert physicians. This module involves inspecting a 3D radiograph slice by slice from the axial, coronal, and sagittal views to extract radiological features. The features extracted from these three views are then integrated for lesion detection and subsequent segmentation. This divide-and-conquer method is data efficient, leveraging the information from numerous 2D radiograph slices to ensure comprehensive lesion feature extraction, which provides a solid foundation for the overall disease diagnosis. Second, a cross-stage module is proposed to facilitate the transfer of information across different tasks. Specifically, the results from the cross-view module are used to identify the slice indices that contain lesions and the detailed lesion locations within those slices. This location information is then utilized in the cross-scale and cross-slice modules to extract the previously mentioned local-scale, global-scale, and 3D-scale lesion features, enabling targeted lesion analysis. Third, a cross-modality module is developed to promote the sharing of knowledge across additional radiographic modalities. The S2S AI is initially developed on these additional modalities and then refined for the target modality. This process enhances the diagnostic performance of the S2S AI by harnessing the rich disease knowledge available from additional data sources.

Data overview

We used a real-world, large-scale, multi-center CT scan cohort of mediastinal neoplasms to develop and evaluate our S2S AI system called S2S-Med in this study. This cohort was sourced from a previous study⁷(see Table 1) and included data from 24 medical centers across China. In contrast to the previous study, we additionally retrieved subtype diagnosis records from these centers and excluded 4 centers due to unavailable records. Of the remaining 20 centers, 15 were allocated as development centers for creating the S2S-Med system, while the other 5 served as testing centers to assess the system's performance (see supplemental information for details of data partitioning). There was no overlap in patient data between the development and testing centers.

plain CT scans from the testing centers. Each scan was accompanied by manually annotated segmentation masks of mediastinal neoplasms prepared by expert radiologists. From the development centers, all contrast CT scans were used as the pretraining set for the cross-modality module, while the plain CT scans were split into a training set (884 plain CT scans) and a validation set (222 plain CT scans) for training and selecting AI models. The plain CT scans from the testing centers were set aside as the test set to evaluate the generalization capability of the AI models.

In the coarse-grained and fine-grained disease classification tasks, we utilized all plain CT scans obtained through the data selection pipeline. The labels for coarse-grained disease types and fine-grained disease subtypes were derived from pathology biopsy results. To ensure extensive coverage of various disease subtypes, we used all data from the development centers for training AI models (4,094 plain CT scans) and reserved data from the testing centers as the test set (1,126 plain CT scans). To make the most of the data that lacked manually annotated segmentation masks of mediastinal neoplasms, we applied AI models from the disease segmentation task to perform slice-wise segmentation on these data.

S2S AI accurately detects mediastinal neoplasms

Mediastinal neoplasms, a subset of thoracic cancers, have experienced an increase in incidence in recent decades. The detection of these lesions is critical for subsequent diagnostic procedures and personalized treatment planning. Although plain CT scans are the primary tool for detection, the complex mediastinal anatomy and the similarity between lesions and surrounding tissue can lead to misdiagnosis by physicians. Therefore, there is an urgent need for a detection method with high sensitivity to mediastinal neoplasms. S2S-Med was developed to meet this need, utilizing both pretraining and training sets, with the best-performing AI models on the validation set selected for evaluation. The detection sensitivity (see methods) was assessed on the test set to underscore the potential of S2S-Med in reducing misdiagnosis.

S2S-Med demonstrated exceptional performance in detecting mediastinal neoplasms from plain CT scans (sensitivity: 0.996, 95% confidence interval [CI]: 0.981–1.000). It successfully identified all lesions (sensitivity: 1.000 [95% CI: 1.000–1.000]) within

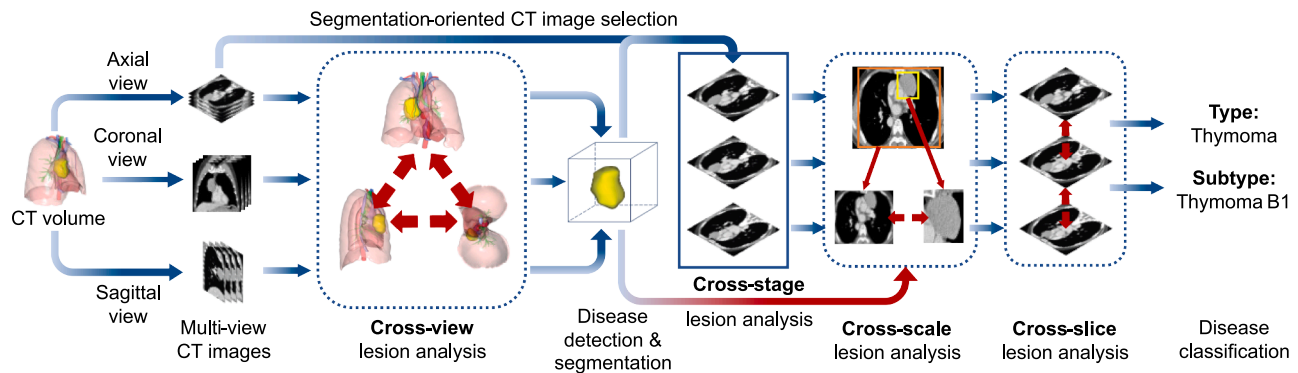


Figure 2. The architecture of S2S AI

11 of 12 coarse-grained disease types and 24 of 25 fine-grained subtypes. Notably, this included all thymomas, which are of great clinical significance; all benign cysts, a common type of mediastinal neoplasm; and all malignant tumors, the misdiagnosis of which could lead to adverse outcomes. These results highlighted the clinical utility of S2S-Med in reducing the risk of misdiagnosis. Detailed results are provided in [Tables S1](#) and [S2](#).

The effectiveness of the cross-view and cross-modality modules was also evaluated in this task. The cross-view module was designed to extract rich context image features from CT scans, while the cross-modality module facilitated knowledge sharing from additional data sources. Both modules were expected to enhance disease detection performance and were thus integrated into S2S-Med. To assess their individual contributions, two ablation models were developed: non-CM S2S-Med, which excluded the cross-modality module, and non-CV S2S-Med, which omitted the cross-view module. All other aspects of these models remained consistent with S2S-Med. The results confirmed that the inclusion of richer context information and additional lesion knowledge significantly improved detection sensitivity. S2S-Med outperformed both ablation models in overall detection sensitivity (0.996 [95% CI: 0.981–1.000] for S2S-Med vs. 0.977 [95% CI: 0.954–0.995] for non-CM S2S-Med, $p = 0.0127$; 0.996 [95% CI: 0.981–1.000] for S2S-Med vs. 0.973 [95% CI: 0.951–0.991] for non-CV S2S-Med, $p = 0.0072$). When examining individual disease types (see [Table S1](#)), the ablation models were found to have misdiagnosed cases of the two most common types, thymomas (S2S-Med: 1.000 [95% CI: 1.000–1.000], non-CM S2S-Med: 0.986 [95% CI: 0.972–1.000], non-CV S2S-Med: 0.972 [95% CI: 0.958–1.000]) and benign cysts (S2S-Med: 1.000 [95% CI: 1.000–1.000], non-CM S2S-Med: 0.986 [95% CI: 0.959–1.000], non-CV S2S-Med: 0.973 [95% CI: 0.946–1.000]), as well as some malignant tumors, such as neuroendocrine tumors (S2S-Med: 1.000 [95% CI: 1.000–1.000], non-CM S2S-Med: 0.857 [95% CI: 0.714–1.000], non-CV S2S-Med: 0.857 [95% CI: 0.714–1.000]). These findings indicate that both modules are essential for accurate detection, particularly in cases of high clinical significance or risk.

S2S AI accurately segments mediastinal neoplasms

The precise delineation of mediastinal neoplasms is essential, as it provides crucial insights into their size and spatial extent, which

are key factors in assessing the nature and severity of these tumors. Consequently, S2S-Med was developed and evaluated for its ability to accurately segment lesion boundaries, following a protocol similar to that used for detection. The segmentation performance on the test set was quantified using the Dice score (see [methods](#)). Additionally, the ablation models, non-CM S2S-Med and non-CV S2S-Med, were similarly evaluated to assess their segmentation capabilities.

S2S-Med demonstrated proficient segmentation of mediastinal neoplasms (Dice score: 0.804, 95% CI: 0.780–0.829). Category-wise segmentation performance is depicted in [Figure 4A](#). S2S-Med excelled in segmenting the two most common disease types: thymoma (0.842 [95% CI: 0.807–0.888]) and benign cyst (0.789 [95% CI: 0.734–0.820]). It also effectively delineated the boundaries of malignant tumors, which can have indistinct margins relative to surrounding normal tissue, including thymic carcinoma (0.827 [95% CI: 0.763–0.854]), lymphoma (0.845 [95% CI: 0.792–0.891]), and neuroendocrine tumor (0.683 [95% CI: 0.600–0.797]). Tumors situated within the complex anatomical structures of the mediastinal visceral compartments, such as lymphadenosis, were also well segmented (0.658 [95% CI: 0.547–0.768]). These findings confirm the ability of S2S-Med to accurately quantify the size of mediastinal neoplasms, providing a robust basis for subsequent stages of diagnosis.

A comparison with CAIMEN-CL,⁷ a leading AI diagnostic system for mediastinal neoplasms, revealed that S2S-Med outperformed CAIMEN-CL in the segmentation task (see [Figure 4B](#), 0.804 [95% CI: 0.780–0.829] for S2S-Med vs. 0.766 [95% CI: 0.738–0.795] for CAIMEN-CL, $p = 0.001$), underscoring its enhanced clinical utility.

The contribution of comprehensive context information and knowledge transfer from additional data sources was also significant in the segmentation task. S2S-Med consistently outperformed both ablation models in overall segmentation performance (see [Figure 4B](#), 0.804 [95% CI: 0.780–0.829] for S2S-Med vs. 0.725 [95% CI: 0.697–0.756] for non-CM S2S-Med, $p < 0.001$; 0.804 [95% CI: 0.780–0.829] for S2S-Med vs. 0.765 [95% CI: 0.737–0.793] for non-CV S2S-Med, $p < 0.001$). Notably, S2S-Med achieved the most substantial improvement over non-CM S2S-Med for neuroendocrine tumors, a typical malignant tumor with limited available samples (0.683 [95% CI: 0.600–0.797] for S2S-Med vs. 0.506 [95% CI: 0.305–0.699] for non-CM S2S-Med, $p < 0.001$), highlighting the role of the

Table 1. Demographic information of the patients and mediastinal neoplasm samples (n = 7,765)

Patients' age (95% CI)			48.9 (48.6–49.3)		
Sex, male-to-female ratio			50.02–49.98		
Mediastinal neoplasm type	Mediastinal neoplasm subtype	No. of samples (%)	Mediastinal neoplasm type	Mediastinal neoplasm subtype	No. of samples (%)
Thymoma	thymoma type AB	798 (10.28)	lymphoma	B cell lymphoma	4 (0.05)
	thymoma type A	185 (2.38)		mucosa-associated lymphoid tissue lymphoma	40 (0.52)
	thymoma type B1	284 (3.66)		T cell lymphoma	24 (0.31)
	thymoma type B2	522 (6.72)		mediastinal large B cell lymphoma	72 (0.93)
	thymoma type B3	339 (4.37)		Hodgkin lymphoma	69 (0.89)
	microscopic thymoma with lymphoid stroma	32 (0.41)		non-Hodgkin lymphoma	53 (0.68)
	metaplastic thymoma	12 (0.15)			
	micronodular thymoma	11 (0.14)			
			ectopic thyroid gland	–	96 (1.24)
			granulomatous inflammation	tuberculosis	72 (0.93)
Benign cyst	–	1,820 (23.44)	neurogenic tumor	paraganglioma	5 (0.06)
Thymic carcinoma	–	521 (6.71)		neuroblastoma	9 (0.12)
Germinoma	yolk sac tumor	29 (0.37)		schwannoma	589 (7.59)
	mature teratoma	369 (4.75)		ganglioneuroma	174 (2.24)
Neuroendocrine tumor	immature teratoma	18 (0.23)	other soft tissue tumor	invasive fibromatosis	6 (0.08)
	germ cell tumor	24 (0.31)		solitary fibrous tumor	53 (0.68)
	atypical carcinoid	41 (0.53)		leiomyoma	9 (0.12)
	typical carcinoid	4 (0.05)		lymphangioma	15 (0.19)
	large cell neuroendocrine carcinoma	8 (0.10)		synovial sarcoma	10 (0.13)
	small cell carcinoma	40 (0.52)			
Thymic hyperplasia	–	326 (4.20)		sarcoma	30 (0.39)
				lipoma	68 (0.88)
Lymph adenosis	Castleman lymph node disease	55 (0.71)		liposarcoma	15 (0.19)
	nodular disease	36 (0.46)		hemangioma	8 (0.10)
				vascular tumor	81 (1.04)
Metastasis	–	117 (1.51)	–	–	–
Type unknown	–	317 (4.08)	subtype unknown	–	355 (4.57)

cross-modality module in enhancing performance on rare disease types through data leverage. Similarly, the greatest improvement over non-CV S2S-Med was seen for lymph adenosis, a typical tumor located in the visceral compartments of the mediastinum, where the anatomic structures are extremely complex (0.658 [95% CI: 0.547–0.768] for S2S-Med vs. 0.454 [95% CI: 0.188–0.720] for non-CV S2S-Med, $p < 0.001$), emphasizing the effectiveness of the cross-view module in extracting contextual structure information for lesion segmentation (see [supplemental information](#) for more detailed ablation analysis).

The cross-view module also facilitated the development of a data-efficient AI model. S2S-Med was compared with CV-3D S2S-Med, which employed a 3D convolutional neural network in place of the cross-view module, across various training sample sizes. For a fair comparison, CV-3D S2S-Med was also equipped with the cross-modality module. S2S-Med consistently outperformed CV-3D S2S-Med, with the performance gap widening as the training set size decreased. Even with the

full dataset, S2S-Med maintained superior performance (0.804 [95% CI: 0.780–0.829] for S2S-Med vs. 0.788 [95% CI: 0.764–0.812] for CV-3D S2S-Med, $p = 0.1123$). This superiority can be attributed to the cross-view module's divide-and-conquer approach, which simplifies the learning task for each model, enabling high segmentation accuracy even with relatively limited data.

S2S AI accurately classifies mediastinal neoplasms

The intricate taxonomy of mediastinal neoplasms, encompassing 12 coarse-grained disease types and up to 43 fine-grained subtypes, presents a significant challenge for accurate diagnosis based solely on medical imaging due to the high degree of inter-category homogeneity and intra-category heterogeneity. To address this challenge, we developed S2S-Med and evaluated its ability to classify both coarse-grained and fine-grained disease categories. Classification accuracy (see [methods](#)) was assessed on the test set for each task.

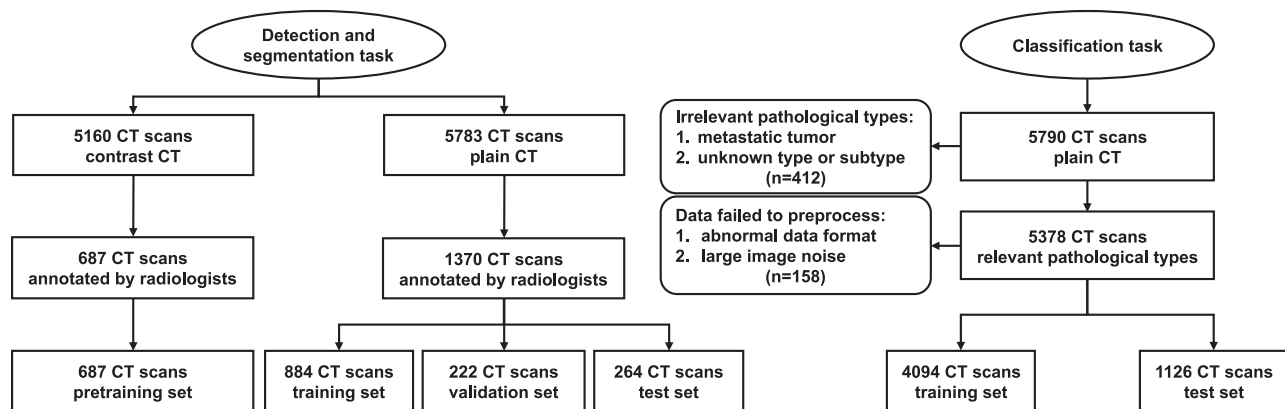


Figure 3. The data selection process of the cohort

S2S-Med exhibited commendable performance in classifying both coarse-grained (accuracy: 0.631, 95% CI: 0.602–0.659) and fine-grained (0.456 [95% CI: 0.426–0.487]) disease subtypes. It achieved a particularly high accuracy rate for thymoma (0.722 [95% CI: 0.674–0.765]), underscoring its clinical significance in diagnosing a disease of considerable importance (see [supplemental information](#) for detailed error analysis). We also compared S2S-Med with CAIMEN-CL in the classification tasks. S2S-Med outperformed CAIMEN-CL in both coarse-grained (0.631 [95% CI: 0.602–0.659] for S2S-Med vs. 0.620 [95% CI: 0.592–0.647] for CAIMEN-CL, $p = 0.1888$) and fine-grained (0.456 [95% CI: 0.426–0.487] for S2S-Med vs. 0.440 [95% CI: 0.410–0.472] for CAIMEN-CL, $p = 0.090$) disease subtypes (see [Figures 5C and 5F](#)), with a significant improvement in thymoma diagnosis (0.722 [95% CI: 0.674–0.765] for S2S-Med vs. 0.671 [95% CI: 0.620–0.720] for CAIMEN-CL, $p = 0.009$), demonstrating the superior performance of S2S-Med.

Precisely locating disease sites and segmenting disease boundaries are crucial for accurate disease classification. The cross-stage module was applied in the classification tasks, which leveraged the results from the disease detection and segmentation tasks to facilitate disease localization and enhance diagnostic accuracy. To illustrate its significance, we developed two ablation models with the same configurations as S2S-Med but with varying degrees of access to segmentation results. The first model, referred to as no-loc S2S-Med, had no access to any detection or segmentation information and could not determine the slice locations of the lesions. The second model, called part-loc S2S-Med, had knowledge about detection information of the slice locations of the lesions but lacked segmentation information about the specific lesion positions within those slices. Results (see [Figures 5B and 5E](#)) showed that the more information that was available, the higher the diagnostic accuracy AI could achieve (0.631 [95% CI: 0.602–0.659] for S2S-Med vs. 0.442 [95% CI: 0.416–0.471] for no-loc S2S-Med, $p < 0.001$; 0.631 [95% CI: 0.602–0.659] for S2S-Med vs. 0.612 [95% CI: 0.582–0.639] for part-loc S2S-Med, $p = 0.061$). Furthermore, in the fine-grained disease classification task, where AI requires more detailed information to analyze fine-grained disease characteristics, precisely locating becomes even more critical (0.456 [95% CI: 0.426–0.487] for S2S-Med vs. 0.266 [95% CI: 0.240–

0.291] for no-loc S2S-Med, $p < 0.001$; 0.456 [95% CI: 0.426–0.487] for S2S-Med vs. 0.412 [95% CI: 0.380–0.442] for part-loc S2S-Med, $p = 0.001$). These results underscore the effectiveness of incorporating diagnostic results from prior steps to enhance subsequent performance.

Lesion characteristics are also pivotal for accurate classification. The cross-scale and cross-slice modules were designed to extract features at global, local, and 3D scales. To assess their contributions, we developed two additional ablation models: non-scale S2S-Med and non-slice S2S-Med, which omitted the cross-scale and cross-slice modules, respectively. Evaluations revealed that S2S-Med's performance was superior, with the largest accuracy gains observed in the fine-grained classification task (0.456 [95% CI: 0.426–0.487] for S2S-Med vs. 0.437 [95% CI: 0.409–0.466] for non-scale S2S-Med, $p = 0.031$; 0.456 [95% CI: 0.426–0.487] for S2S-Med vs. 0.439 [95% CI: 0.411–0.471] for non-slice S2S-Med, $p = 0.052$), highlighting the importance of detailed lesion feature extraction for precise diagnosis.

S2S AI assists humans positively

We conducted a human-AI experiment to assess the assisting capabilities of S2S-Med in real-world clinical settings. We recruited 9 physicians from Guizhou Provincial People's Hospital, representing senior, mid-level, and junior groups based on their qualifications. We randomly selected 2 CT scans for each fine-grained disease subtype from the test set, forming the human-AI validation set comprising 86 CT scans. In the experiment, the physicians first independently reviewed these CT scans and determined the coarse-grained disease types and fine-grained disease subtypes of mediastinal neoplasms depicted in the scans. Then, the predictions of S2S-Med were presented to them, and they were required to determine the disease types and subtypes again. Diagnostic accuracies were evaluated based on the diagnoses made by the physicians themselves and those made after reviewing the predictions of S2S-Med.

The involvement of S2S-Med significantly enhanced physicians' diagnostic accuracy (see [Figures 6A and 6D](#)), and the improvement was more pronounced when differentiating the fine-grained disease subtypes, indicating the clinical value of S2S-Med in assisting physicians with complex diseases. However, although S2S-Med successfully assisted each physician

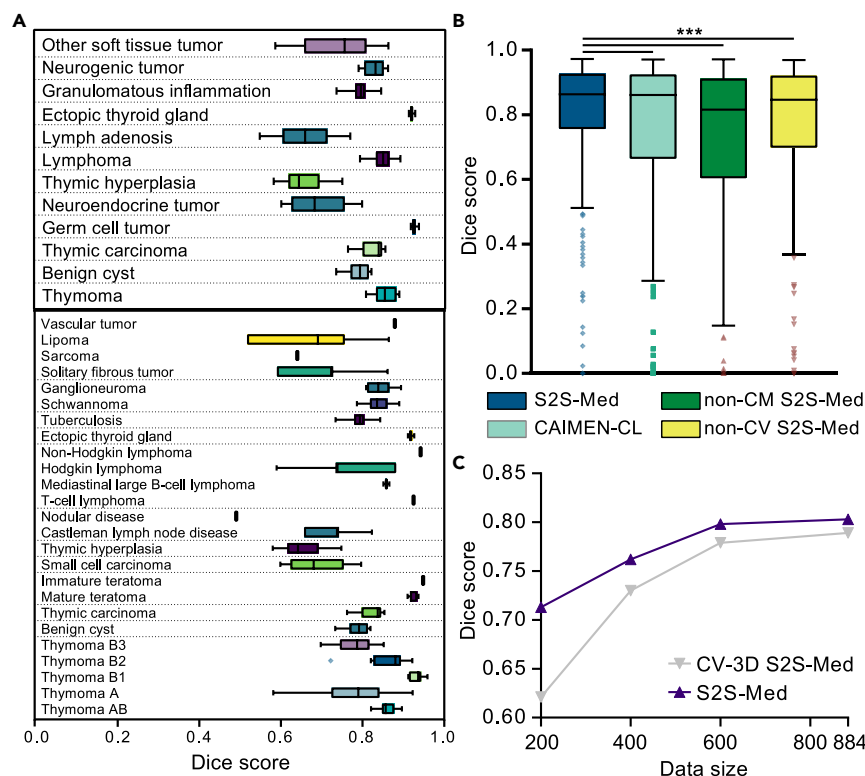


Figure 4. S2S AI accurately segments mediastinal neoplasms

(A) Category-wise Dice scores of S2S-Med on the test set.

(B) Dice scores of S2S-Med, CAIMEN-CL, non-CM S2S-Med, and non-CV S2S-Med on the test set.

(C) Dice scores on the test set of S2S-Med and CV-3D S2S-Med trained with different data sizes. See also [Tables S3](#) and [S4](#). Error bars in the boxplot elements represent the Tukey confidence interval.

DISCUSSION

With the advancements in diagnostic radiology, radiological diagnosis has taken on an increasingly pivotal role in disease management. Therefore, it is essential to ensure accurate decisions for optimal patient care. To our knowledge, we are the first to delve into the radiological diagnosis of fine-grained diseases by proposing the S2S AI. We have further validated the effectiveness of S2S AI by developing an S2S-Med system using a large-scale real-world CT scan cohort of mediastinal neoplasms. The

high diagnostic accuracy demonstrated by S2S-Med underscores the potential of applying S2S AI to other complex diseases. We have also evaluated the capacity of S2S AI on a dataset of kidney tumors, which also presents satisfactory results (see [supplemental information](#) for further details).

The S2S AI pipeline is designed with flexibility to accommodate the radiological diagnosis for various diseases that may have different characteristics, including radiographic modalities and disease granularities. In the task of disease segmentation, for 2D radiographs, the cross-view module can utilize three models developed on different splits of the 2D radiograph dataset. These models are then combined using an ensemble approach to make predictions. In the classification task, the 3D scale module enables adjusting the covering range along the image channels to accommodate radiographs with a varying number of radiograph slices.

The initial critical step of radiological diagnosis is the comprehensive identification and precise segmentation of lesions. In the case of mediastinal neoplasms, S2S-Med achieves high detection sensitivity on the test set through the implementation of the cross-view module, thereby demonstrating the effectiveness of this proposed approach in minimizing misdiagnosis. Moreover, S2S-Med achieves a high Dice score on the test set, indicating satisfactory performance in accurately quantifying lesion sizes.

Another crucial factor in determining the treatment strategy is the classification of disease categories, which includes both coarse-grained disease types and fine-grained disease subtypes. In clinical practice, diseases such as cancers are typically diagnosed through biopsy.⁸ S2S-Med, which diagnoses

in improving their diagnostic accuracy, no physician's performance after assistance could surpass that of S2S-Med. These findings suggested that for complex diseases such as mediastinal neoplasms, manual diagnosis based on radiographs could be challenging.

We also observed heterogeneous AI assisting effects on different physicians (see [Figures 6C](#) and [6F](#)). We defined physicians' trust rate in S2S-Med, which was represented as the percentage of samples where the diagnoses of physicians and S2S-Med initially differed, and physicians subsequently changed their predictions to align with those of S2S-Med. In the experiment, all physicians demonstrated positive trust rates, but these trust rates were not strongly correlated with physicians' diagnostic expertise without the assistance of S2S-Med ($R^2 = 0.0189$ for coarse-grained disease type classification and $R^2 = 0.0001$ for fine-grained disease subtype classification). Despite physicians' varying rates of trust in S2S-Med, we observed that the higher the trust rate was, the higher the diagnostic accuracy physicians could achieve after S2S-Med assistance (see [Figures 6B](#) and [6E](#)). We analyzed the linear correlation between physicians' trust rates in S2S-Med and their diagnostic accuracy with the assistance of S2S-Med, and observed a positive correlation between the two factors in both coarse-grained ($R^2 = 0.3554$) and fine-grained ($R^2 = 0.7446$) classifications of mediastinal neoplasms. These results indicated that S2S-Med was worthy of trust for physicians. Additionally, the stronger correlation between the two factors in fine-grained disease subtype classification compared to coarse-grained disease type classification may suggest that for more challenging tasks, greater trust in AI could lead to higher diagnostic accuracy.

diseases using pre-operative CT scans, can provide early warnings for specific diseases and help to avoid unforeseen harm to patients during biopsy or other invasive procedures. Therefore, S2S AI has the potential to be applied in medical practices to facilitate personalized medicine for patients.

Several studies have been conducted on the multi-stage radiological diagnosis of various diseases. Zhou et al.⁹ investigated the AI-based segmentation and classification of breast lesions using enhanced MRI. Zhao et al.¹⁰ developed a comprehensive AI system for the segmentation and classification of diabetic retinopathy and branch retinal vein occlusion using fundus fluorescein angiography images. Tavakoli et al.¹¹ employed AI for the segmentation and classification of white blood cells in peripheral smear images. Son et al.¹² proposed an AI-based method for the localization and classification of skin diseases with erythema using skin images. However, these studies primarily cover coarse-grained disease types with no more than 5 subtypes. In contrast, our S2S-Med system demonstrates satisfactory performance in classifying 43 fine-grained disease subtypes, indicating its potential for broader application across various diseases.

Human-AI collaboration in medicine has garnered increasing attention from researchers in recent years, and multiple studies have revealed that AI assistance often has different effects on different individual physicians or groups.^{13,14} In this study, we explore the human-AI collaboration on complex fine-grained diseases and we also observe that S2S-Med has heterogeneous effects on physicians with varying expertise and on different diagnostic difficulties. Overall, the more that physicians trust S2S-Med, the higher the diagnostic accuracy they can achieve with the assistance of S2S-Med. Moreover, when diagnosing fine-grained disease subtypes that are more challenging than diagnosing coarse-grained disease types, these trends become even more pronounced. However, we further observed that physicians' trust rates in S2S-Med were not significantly related to their diagnostic expertise. These results suggest that in the diagnosis of complex diseases through human-AI cooperation, the trust that physicians place in AI is a critical factor. When designing human-AI cooperation protocols, it is essential to not only focus on the technical capabilities of the AI system but also consider how to establish and reinforce the trust of physicians in the AI system.

Although S2S-Med's performance surpasses that of physicians, and trust in S2S-Med can lead to increased diagnostic accuracy for physicians, overreliance on S2S-Med could introduce the risk of misdiagnosing cases that were correctly identified by physicians. In the classification tasks, physicians with a high trust rate in S2S-Med could produce up to one "negative" case, indicating a misdiagnosis that the AI system did not identify. This underscores the importance of maintaining a balance between reliance on AI and physician confidence.

Despite the satisfactory accuracy for diagnosis, the deployment of our S2S AI system in diverse clinical settings presents several challenges that warrant careful consideration. First, many clinical settings lack dedicated GPU servers for AI deployment. A possible solution is to develop cloud-based solutions with secure architectures. Second, potential overreliance on AI by clinicians could impact diagnostic decision-making. A potential solution is to implement interpretable AI interfaces and clinician training programs.

In conclusion, this study introduces S2S AI as a means to enhance the radiological diagnosis of fine-grained diseases. The proposed S2S AI system has been validated using a large-scale CT scan cohort of mediastinal neoplasms, which encompasses 43 disease types. The system demonstrates high diagnostic accuracy in tasks such as lesion detection, segmentation, and classification. The satisfactory diagnostic accuracy of the S2S AI system suggests its potential for broader application in medical practice to address other complex diseases. The insights from the human-AI collaboration experiment may also contribute to new perspectives on the collaboration between humans and AI for the diagnosis of complex diseases.

METHODS

Data preprocess

Prior to utilizing the CT scans for segmentation and classification tasks, we conducted preprocessing to eliminate non-mediastinum regions that could introduce interference. More specifically, we utilized a DenseNet121 network that had been trained with lung lobe segmentation annotations to extract a bounding box encompassing the lung areas (see [supplemental information](#) for details of DenseNet121 training and bounding box selection). Subsequently, we eliminated non-relevant regions located outside this bounding box. It is important to note that since the mediastinum areas aligned with the lung areas, this preprocessing approach did not impact the diagnostic performance of mediastinal neoplasms. Additionally, to ensure consistency, all CT scans and their corresponding ground truth segmentation masks were resized to a resolution of $256 \times 256 \times 256$ (see [supplemental information](#) for details about resolution selection).

The model architecture in the detection and segmentation tasks

In the detection and segmentation tasks, we selected nnUNet¹⁵ as our baseline model due to its excellent performance across various medical segmentation tasks. Both the 2D and 3D versions of nnUNet were utilized. The configurations provided by nnUNet, based on the characteristics of our dataset, were directly applied (see [supplemental information](#) for detailed configurations).

For the 2D nnUNet approach, individual 2D CT slices were extracted along the axial, coronal, or sagittal axes from the 3D CT scans. These slices were then input into the 2D nnUNet to generate corresponding 2D binary segmentation masks. These masks consisted of values of 0 or 1, indicating the presence or absence of mediastinal neoplasms, respectively.

For the 3D nnUNet approach, the entire 3D CT scans were fed directly into the 3D nnUNet. This allowed for the prediction of corresponding 3D binary segmentation masks specifically for mediastinal neoplasms.

The cross-view module. The cross-view module entailed the development of three 2D nnUNets that collectively performed segmentation mask prediction. Initially, all CT scans in the training set were segmented into a series of 2D CT slices along the axial, coronal, and sagittal axes, yielding three distinct cross-view training sets. Subsequently, individual 2D nnUNets were trained on each specific cross-view training set using a

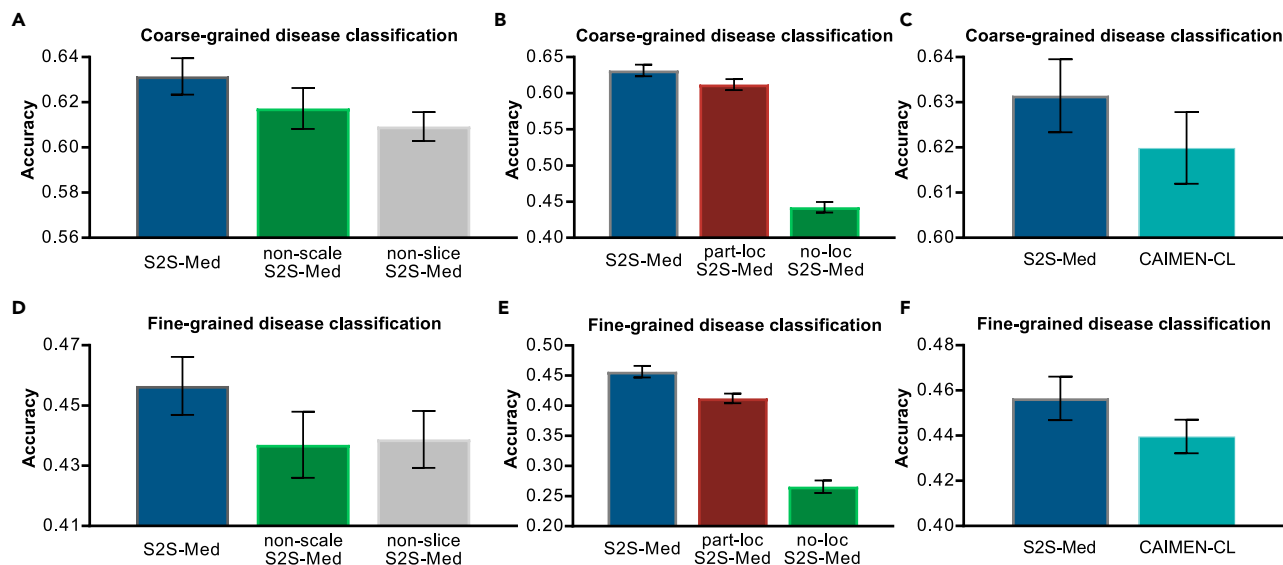


Figure 5. S2S AI accurately classifies mediastinal neoplasms

(A) The coarse-grained classification accuracy of S2S-Med, non-scale S2S-Med, and non-slice S2S-Med on the test set.
 (B) The coarse-grained classification accuracy of S2S-Med, part-loc S2S-Med, and no-loc S2S-Med on the test set.
 (C) The coarse-grained classification accuracy of S2S-Med and CAIMEN-CL on the test set.
 (D) The fine-grained classification accuracy of S2S-Med, non-scale S2S-Med, and non-slice S2S-Med on the test set.
 (E) The fine-grained classification accuracy of S2S-Med, part-loc S2S-Med, and no-loc S2S-Med on the test set.
 (F) The fine-grained classification accuracy of S2S-Med and CAIMEN-CL on the test set.

Error bars in the boxplot elements represent the Tukey confidence interval.

supervised loss function that compared the predicted and ground truth segmentation masks.

During the testing phase, each CT scan in the test set was segmented into three sets of 2D CT slices along the axial, coronal, and sagittal axes, respectively. The corresponding trained 2D nnUNet was then utilized to predict segmentation masks for the 2D CT slices, resulting in three sets of 2D segmentation masks. These 2D segmentation masks within each set were combined to create a single 3D segmentation mask. To generate the final 3D segmentation mask, the three individual 3D segmentation masks were integrated using a voxel-wise logical operation (see [supplemental information](#) for details). In other words, a voxel in the final 3D segmentation mask would have a value of 1 if any of the corresponding voxels in the three individual 3D segmentation masks had a value of 1.

The cross-modality module. In both the detection and segmentation tasks, the cross-modality module was employed. Initially, the models were trained on the pretraining set. Subsequently, the pretrained models were fine-tuned on the training set without freezing any parameters. For practical implementation, the cross-modality module was implemented using the application programming interfaces of nnUNet (see [supplemental information](#) for further details).

The model architecture in the classification task

For the classification task, we employed ResNet50¹⁶ as the baseline model. To process a 3D CT scan, we initially extracted 10 representative 2D CT slices along with their corresponding 2D segmentation masks. These slices were then combined to create a 3D CT sub-scan and a 3D segmentation sub-mask, both with

resolutions of $10 \times 256 \times 256$ (depth $\times$ height $\times$ width). Subsequently, we input the 3D CT sub-scan and the 3D segmentation sub-mask into the model for the purpose of conducting coarse-grained or fine-grained disease type classification.

The cross-scale module. In the cross-scale module, we employed a multi-head attention mechanism¹⁷ to integrate global-scale lesion features with local-scale lesion features. Specifically, we utilized the CT feature obtained from ResNet50, which had a shape of $10 \times 256 \times 8 \times 8$, as the global-scale feature. Concurrently, we resized the segmentation mask to 8×8 and extracted a bounding box containing mediastinal neoplasms within this smaller scale. Then, we performed a region of interest (ROI) pooling operation on the CT feature using the extracted bounding box. This yielded a local-scale feature with a shape of $10 \times 256 \times 2 \times 2$.

The local-scale feature was input into a 2D convolutional layer to generate the query feature with a shape of 10×256 . Additionally, the global-scale feature was flattened and passed through two individual linear layers to obtain the key and value features, both with shapes of $10 \times 16,384$. We then extracted 10 one-dimensional (1D) query, key, and value vectors along the depth dimension and utilized them to compute multi-head attention with two heads. This computation resulted in 10 global-local vectors. Subsequently, we averaged the 10 global-local vectors to obtain a cross-scale feature with a shape of 1×256 .

The cross-slice module. In the cross-slice module, we employed the multi-head attention mechanism to fuse the features of each CT slice with those of adjacent slices. Similar to the cross-scale module, we utilized the CT feature extracted by ResNet50, which had a shape of $10 \times 256 \times 8 \times 8$.

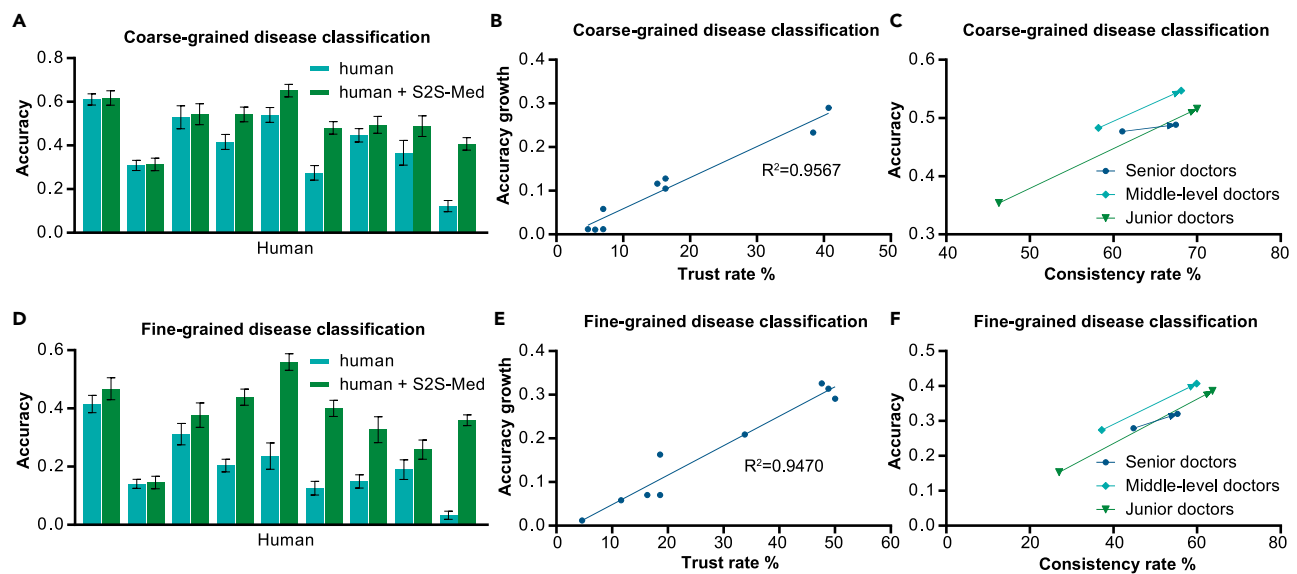


Figure 6. S2S AI delivers positive assistance to physicians in the human-AI experiment

(A) The coarse-grained classification accuracy of physicians without/with the assistance of S2S-Med. The horizontal dotted line represented the accuracy of S2S-Med.
 (B) The physicians' trust rates in S2S-Med and their diagnostic accuracy without AI assistance in the coarse-grained disease-type classification task.
 (C) The physicians' trust rates in S2S-Med and their diagnostic accuracy with AI assistance in the coarse-grained disease-type classification task.
 (D) The fine-grained classification accuracy of physicians without/with the assistance of S2S-Med. The horizontal dotted line represented the accuracy of S2S-Med.
 (E) The physicians' trust rates in S2S-Med and their diagnostic accuracy without AI assistance in the fine-grained disease subtype classification task.
 (F) The physicians' trust rates in S2S-Med and their diagnostic accuracy with AI assistance in the fine-grained disease subtype classification task.
 Error bars in the boxplot elements represent the Tukey confidence interval.

The CT feature was processed through a 2D convolutional layer to generate a query feature with a shape of 10×256 . Additionally, by flattening the CT feature and passing it through two linear layers, we obtained key and value features with shapes of $10 \times 16,384$.

During the attention computation, each 1D query vector extracted from the query feature was matched with a key and value vector. These key and value vectors were obtained by concatenating the adjacent 5 channels of the query vector in the key and value features along the depth dimension, resulting in a shape of $10 \times 81,920$. To ensure valid coverage, for query vectors located at the edge of the query features, the range of adjacent channels was adjusted accordingly. The multi-head attention computation was then performed among the query, key, and value vectors, resulting in 10 cross-3D vectors. Finally, these 10 cross-3D vectors were averaged to produce a cross-slice feature with a shape of 1×256 .

The cross-stage module. The cross-stage module utilized segmentation masks to localize mediastinal neoplasms in the classification tasks. First, when extracting the 3D CT sub-scan, only CT slices with segmented mediastinal neoplasms were selected. In cases where the number of CT slices with segmented mediastinal neoplasms was less than 10, some CT slices were replicated to ensure a total of 10 slices. Second, in the cross-scale module, the segmentation masks were used to retrieve the bounding box of mediastinal neoplasms for the purpose of performing the ROI pooling operation.

For both the no-seg S2S-Med and part-seg S2S-Med approaches, the segmentation masks were replaced with a zero-value mask. During the ROI pooling operation, the bounding box of the mediastinal neoplasms was substituted with a bounding box encompassing the entire content of the CT slice, thereby removing the accurate position information of the mediastinal neoplasms. Moreover, for no-seg S2S-Med, CT slices with or without segmented mediastinal neoplasms were chosen in an equal-interval pattern from the 3D CT scans, meaning that no-seg S2S-Med lacked any coarse position information regarding mediastinal neoplasms.

Evaluation metrics and statistical analysis

In the segmentation task, the performance of S2S-Med was evaluated using the sensitivity and Dice score metrics:

$$\text{Sensitivity} = TP_{\text{scan}} / (TP_{\text{scan}} + FN_{\text{scan}}),$$

where TP_{scan} and FN_{scan} represented the number of true positive and false negative scans. A scan was considered positive when the predicted 3D segmentation mask overlapped with the ground truth 3D segmentation mask.

$$\text{Dice score} = 2TP_{\text{voxel}} / (FP_{\text{voxel}} + 2TP_{\text{voxel}} + FN_{\text{voxel}}),$$

where TP_{voxel} , FP_{voxel} , and FN_{voxel} represented the number of true positive, false positive, and false negative voxels in the 3D CT scan. The Dice score was computed at the scan level by

comparing the predicted 3D segmentation mask with the ground truth 3D segmentation mask.

In the classification tasks, the performance of S2S-Med was evaluated using the accuracy metric:

$$\text{Accuracy} = \#(\text{samples correctly classified}) / \#(\text{all samples}),$$

where # represented the number of samples.

To compare different models, a one-sided Wilcoxon signed rank test¹⁸ was employed for all metrics (see [supplemental information](#) for further details). A $p < 0.05$ was considered statistically significant. Bootstrapping with $n = 1,000$ was used to compute the 95% CIs for all metrics (see [supplemental information](#) for further details).

Training details

All codes were implemented using Python, and the models were developed using PyTorch. For the detection and segmentation tasks, the models were pretrained for 500 epochs and then fine-tuned for 1,000 epochs. The specific training configurations, including the learning rate, batch size, and loss functions, were determined by the nnUNet framework. The models with the highest performance on the validation set were chosen for evaluation.

In the classification task, the models were pretrained for 20 epochs and then fine-tuned for 40 epochs. The learning rate was set to 0.01, and the batch size was set to 12. The cross-entropy loss function, which compared the predicted category with the ground truth category, was utilized. The models at the end of the training process were selected for evaluation.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Feng Xu (feng-xu@tsinghua.edu.cn).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- Deidentified DICOM sequential data and corresponding diagnosis records of 50 patients from the test set have been deposited at Zenodo¹⁹ and are publicly available as of the date of publication at <https://doi.org/10.5281/zenodo.15044407>. Remaining deidentified DICOM sequential data and diagnosis records reported in this paper will be shared by the [lead contact](#) upon request.
- All original code has been deposited at Zenodo and is publicly available at <https://doi.org/10.5281/zenodo.15044407>.

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AUTHOR CONTRIBUTIONS

R.T., Y.G., and F.X. were responsible for the conception and design of this study. R.T. developed the S2S-Med system. Y.G. and F.X. assisted in improving the system. R.T. conducted the experimental analysis. R.T., Y.G., and F.X. wrote the manuscript. All authors read and approved the manuscript. All authors had access to all the raw datasets. R.T. has accessed and verified the data. R.T., Y.G., and F.X. were responsible for the decision to submit the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

SUPPLEMENTAL INFORMATION

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REFERENCES

- Wang, R.C., and Wang, Z. (2023). Precision Medicine: Disease Subtyping and Tailored Treatment. *Cancers* 15, 3837. <https://doi.org/10.3390/cancers15153837>.
- Doi, K. (2006). Diagnostic imaging over the last 50 years: research and development in medical imaging science and technology. *Phys. Med. Biol.* 51, R5–R27. <https://doi.org/10.1088/0031-9155/51/13/R02>.
- Doi, K. (2007). Computer-aided diagnosis in medical imaging: historical review, current status and future potential. *Comput. Med. Imaging Graph.* 31, 198–211. <https://doi.org/10.1016/j.compmedimag.2007.02.002>.
- Kwee, T.C., Almaghrabi, M.T., and Kwee, R.M. (2023). Diagnostic radiology and its future: what do clinicians need and think? *Eur. Radiol.* 33, 9401–9410. <https://doi.org/10.1007/s00330-023-09897-2>.
- Zhou, Z., Zhao, C., Qiao, H., Wang, M., Guo, Y., Wang, Q., Zhang, R., Wu, H., Dong, F., Qi, Z., et al. (2022). RATING: Medical knowledge-guided rheumatoid arthritis assessment from multimodal ultrasound images via deep learning. *Patterns* (New York, N.Y.) 3, 100592. <https://doi.org/10.1016/j.patter.2022.100592>.
- Lyu, Q., Namjoshi, S.V., McTye, E., Topaloglu, U., Barcus, R., Chan, M.D., Cramer, C.K., Debinski, W., Gurcan, M.N., Lesser, G.J., et al. (2022). A transformer-based deep-learning approach for classifying brain metastases into primary organ sites using clinical whole-brain MRI images. *Patterns* (New York, N.Y.) 3, 100613. <https://doi.org/10.1016/j.patter.2022.100613>.
- Tang, R., Liang, H., Guo, Y., Li, Z., Liu, Z., Lin, X., Yan, Z., Liu, J., Xu, X., Shao, W., et al. (2023). Pan-mediastinal neoplasm diagnosis via nationwide federated learning: a multicentre cohort study. *Lancet Digit. Health* 5, e560–e570. [https://doi.org/10.1016/S2589-7500\(23\)00106-1](https://doi.org/10.1016/S2589-7500(23)00106-1).
- Ridout, A.J., Kasivisvanathan, V., Emberton, M., and Moore, C.M. (2014). Role of magnetic resonance imaging in defining a biopsy strategy for detection of prostate cancer. *Int. J. Urol.* 21, 5–11. <https://doi.org/10.1111/iju.12259>.
- Zhou, H., Hua, Z., Gao, J., Lin, F., Chen, Y., Zhang, S., Zheng, T., Wang, Z., Shao, H., Li, W., et al. (2024). Multitask Deep Learning-Based Whole-Process System for Automatic Diagnosis of Breast Lesions and Axillary Lymph Node Metastasis Discrimination from Dynamic Contrast-Enhanced-MRI: A Multicenter Study. *J. Magn. Reson. Imaging* 59, 1710–1722. <https://doi.org/10.1002/jmri.28913>.
- Zhao, X., Lin, Z., Yu, S., Xiao, J., Xie, L., Xu, Y., Tsui, C.K., Cui, K., Zhao, L., Zhang, G., et al. (2023). An artificial intelligence system for the whole

- process from diagnosis to treatment suggestion of ischemic retinal diseases. *Cell Rep. Med.* 4, 101197. <https://doi.org/10.1016/j.xcrm.2023.101197>.
11. Tavakoli, S., Ghaffari, A., Kouzehkhanan, Z.M., and Hosseini, R. (2021). New segmentation and feature extraction algorithm for classification of white blood cells in peripheral smear images. *Sci. Rep.* 11, 19428. <https://doi.org/10.1038/s41598-021-98599-0>.
 12. Son, H.M., Jeon, W., Kim, J., Heo, C.Y., Yoon, H.J., Park, J.U., and Chung, T.M. (2021). AI-based localization and classification of skin disease with erythema. *Sci. Rep.* 11, 5350. <https://doi.org/10.1038/s41598-021-84593-z>.
 13. Yu, F., Moehring, A., Banerjee, O., Salz, T., Agarwal, N., and Rajpurkar, P. (2024). Heterogeneity and predictors of the effects of AI assistance on radiologists. *Nat. Med.* 30, 837–849. <https://doi.org/10.1038/s41591-024-02850-w>.
 14. Groh, M., Badri, O., Daneshjou, R., Koochek, A., Harris, C., Soenksen, L. R., Doraiswamy, P.M., and Picard, R. (2024). Deep learning-aided decision support for diagnosis of skin disease across skin tones. *Nat. Med.* 30, 573–583. <https://doi.org/10.1038/s41591-023-02728-3>.
 15. Isensee, F., Jaeger, P.F., Kohl, S.A.A., Petersen, J., and Maier-Hein, K.H. (2021). nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nat. Methods* 18, 203–211. <https://doi.org/10.1038/s41592-020-01008-z>.
 16. He, K., Zhang, X., Ren, S., and Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR)*, pp. 770–778.
 17. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. (2017). Attention is all you need. *Adv. Neural Inf. Process. Syst.* 30.
 18. Woolson, R.F. (2005). Wilcoxon Signed-Rank Test. In *Encyclopedia of Biostatistics*, P. Armitage and T. Colton, eds. (John Wiley and Sons). <https://doi.org/10.1002/0470011815.b2a15177>.
 19. Gerczuk, M. (2023). HyperPersonalisation. Zenodo. <https://doi.org/10.5281/zenodo.8328092>.